



maxxwell production
well slotting service

MAXXWELL PRODUCTION PROGRAM

MAXIMIZE PRODUCTION
EXTEND WELL LIFE

HYDRO-SLOTTING PERFORATION

OIL WELL BARRACUDA PINTO 03-04-002-03W2 (74G003)

**ECOLOGICALLY SAFE,
ENVIRONMENTALLY
FRIENDLY, EFFECTIVE
AFFORDABLE METHOD**

MAXXWELL PRODUCTION INC.

72 Beechbrooke way,
Aurora, ON, Canada
L4G 6N7

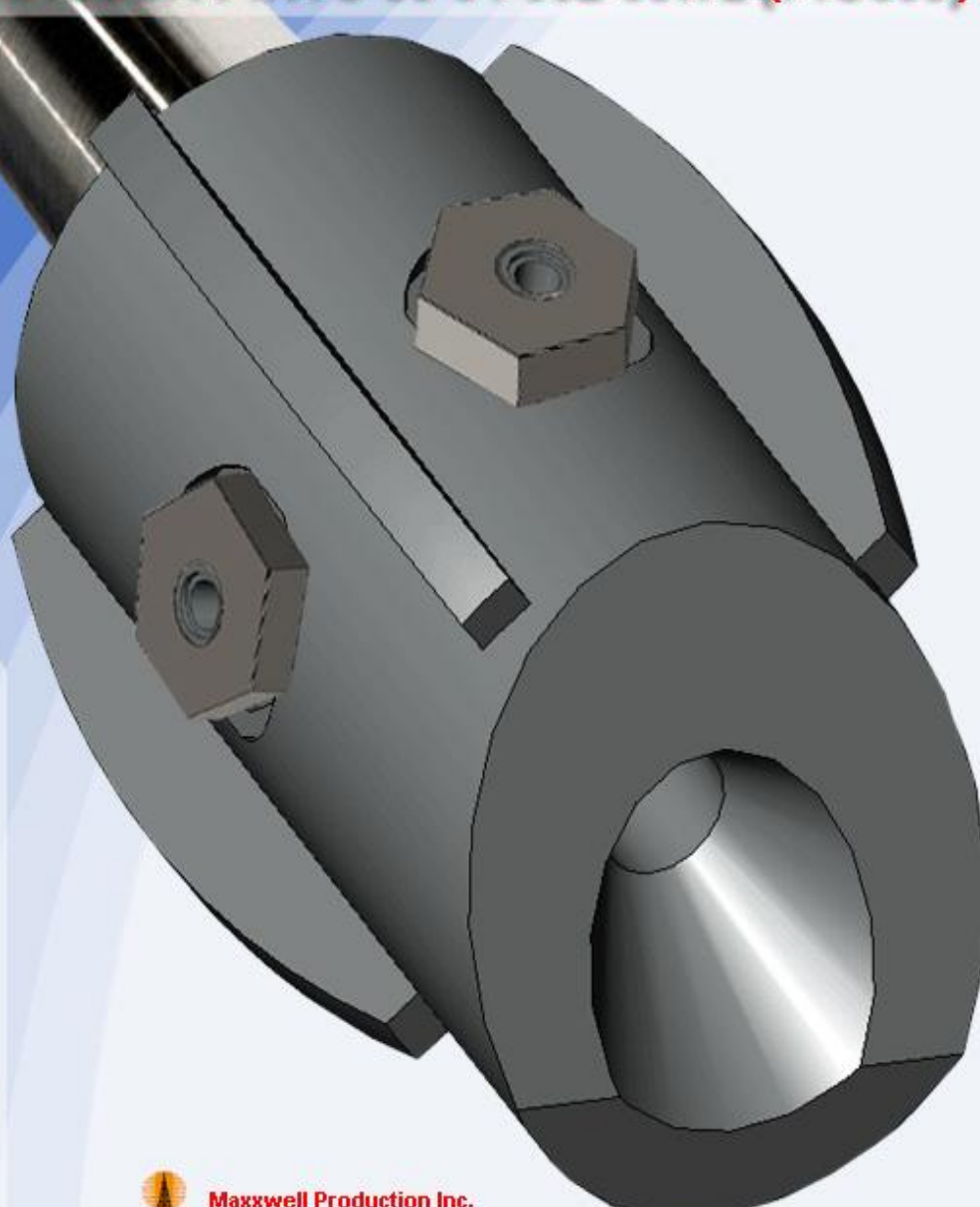
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Maxxwell Production Inc.



Stanford University

(Department of Geophysics)

"Abrasive hydro-slotting perforation is simple, but very effective method of well production enhancement which incorporates modern concepts including the interaction between reservoir mechanics and the physical permeability of reservoir rocks.

Based on the available information, including theoretical calculation, laboratory experiments, and field test results, we believe that this method may find effective application in the oil and mining industries."

Amos Nur

Prof. of Geophysics and Director,
Stanford Rock Physics and Borehole
Geophysics Project
Stanford University

Lev Vernik

Research Associate
Stanford Rock Physics and Borehole
Geophysics Project
Stanford University





December 10, 2013

SK-003

SLOT DISCHARGE OF PRODUCTIVE LAYER WITH HYDRO-SLOTTING PERFORATION

OIL WELL BARRACUDA PINTO **03-04-002-03W2 (74G003)**

MIDALE / FROBISHER FORMATION

Date: December 10, 2013

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Maxxwell Production Inc.
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For: Quinton Hardage, P. Eng.
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OIL AND GAS PRODUCT INTENSIFICATION AND RETRIEVAL TECHNOLOGY

MAXXWELL PRODUCTION INC.
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OIL WELL BARRACUDA PINTO 03-04-002-03W2 (74G003)

Present a comparative analysis of the well prepared on the basis of information provided by the Customer (AOL) :

<http://www.wellslotting.com/318606249670643186062/496706431860624967064/030400203W2.htm>

1. [Map 1 of 03-04-002-03W2 \(74G003\)](#)
2. [Map 2 of 03-04-002-03W2 \(74G003\)](#)
3. [Map 3 of 03-04-002-03W2 \(74G003\)](#)
4. [Map 4 of 03-04-002-03W2 \(74G003\)](#)
5. [Chemical and Geological report \(Water Analysis\) from 26/07/85](#)
6. [Well Ticket from 11/06/13](#)
7. [Log of the well 03-04-002-03W2 \(74G003\)](#)
8. [Production History from 10/1974 - 05/2013](#)
9. [Well Completion Data from 07/20/75](#)
10. [Well side and access road of the well 03-04-002-03W2](#)
11. [Pressure Test from 07/21/74](#)
12. [Application for MPR-Oil well 03-04-002-03W2](#)
13. [Drill Tests from 12/07/74](#)
14. [Geological data sheet from 11/26/74](#)
15. [Oil well report from 4/2/75](#)
16. [Well activity report from 11/01/2005](#)
17. [Well Bulletin for 03-04-002-03W2](#)
18. [Licence to drill a well](#)

WELL DATA

Company	Admiralty
Well	Barracuda Pinto 03-04-002-03W2
Field	Pinto
Province/County	Saskatchewan
Country/State	Canada
Location	101/03-04-002-03W2/00
LSD	03
SEC	04
TWP	002
RGE	03W21
Service	HBC
Licence Number	74G003
Permanent Datum	GL
Coordinates well (Latitude)	49.08955 deg N
Coordinates well (Longitude)	102.35603 deg W
TVD	1584 m (5200 ft)
FM@TD	1584 m (5200 ft)
OD casing	114.3 mm (4.5")
ID casing	103.89 mm (4.09")
Bit Size	200.00 mm
Hole Fluid Type	GEL CHEMICAL
Density/Viscosity	1130.0 kg/m (106.00 sec/L)
PH/Fluid Loss	9.00 (7,50 ml/ 30 min)
Sample Source	FLOW LINE
Rm @ Measured Temp	0.90 @ 20.0 ohm-m
Rmf @ Measured Temp	0.70 @ 16.0 ohm-m



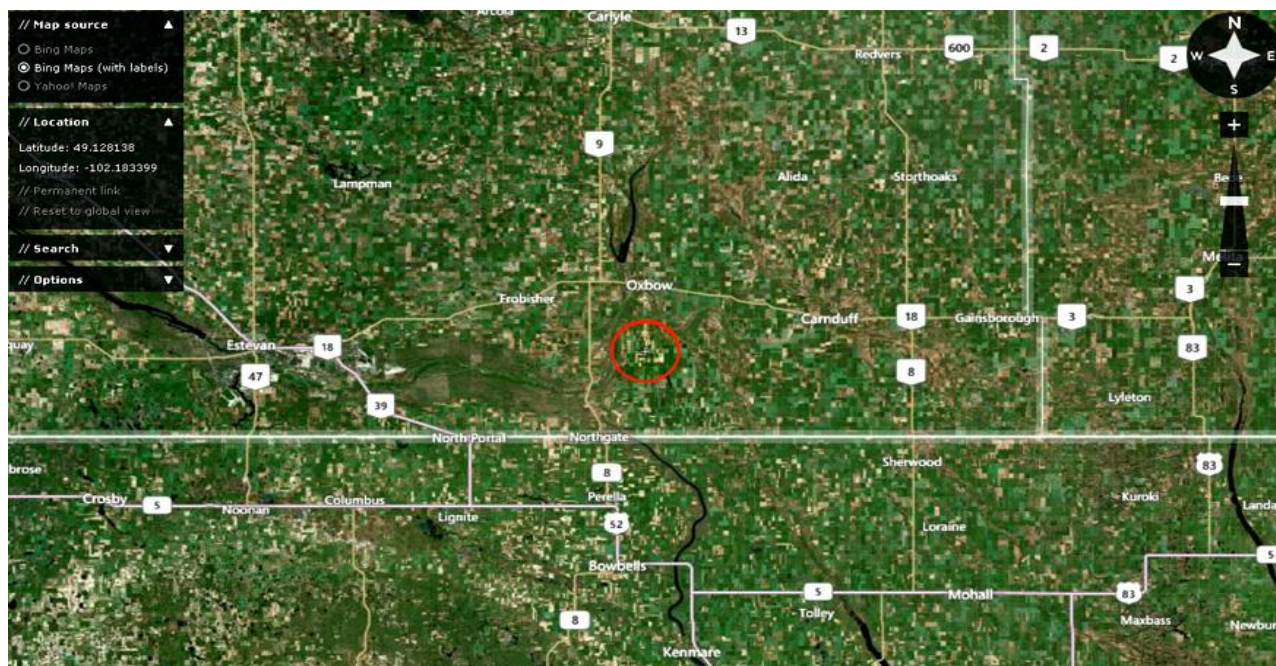
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Rmc @ Measured Temp	0.30 @ 18.0 ohm-m
Source Rmf/Rmc	PRESS
Rm @ BHT	0.81 @ 24.0 ohm-m
Time Since Circulation	3
Max Recorded Temp	57°C (layer level)
Equipment/Base	13052(MDH)
Casing area volume	33.51 m ³
Fluid volume	14.99 m ³





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OPENING FORMATION

Midale /Frobisher	1529.5-1535.0 m and 1536.8-1538.7 m
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SUMMARY ANALYSIS

Summary analysis of geological and geophysical data for the oil well Barracuda Pinto **03-04-002-03W2 (74G003)**

Baseline:

- ▶ Log of the well **03-04-002-03W2**, complex **GIS** analysis, tests results of oil and gas.
- ▶ Geological and geophysical data on nearby wells not represented.
- ▶ When tested in an open hole of the interval **1528.0-1545.0** m (**5013.0 – 5069.0** ft) Midale formation (**21.07.1974**) was obtained a weak flow of oil.
- ▶ When tested in an open hole of the interval **1549.0-1565.7** m (**5082.0 – 5137.0** ft) Frobisher formation (**21.07.1974**) was obtained a weak flow of oil and water.
- ▶ **GIS** data collectors are highlighted in the following ranges:

Top (m)	Bottom (m)	Effective thickness (m)	Saturation of the test results in an open hole
1538.0	1539.5	1.5	Oil
1544.0	1545.3	1.3	Oil + water
1553.5	1554.8	1.3	Oil + water
1562.8	1564.0	1.2	Oil + water
1565.2	1566.6	1.4	Oil + water

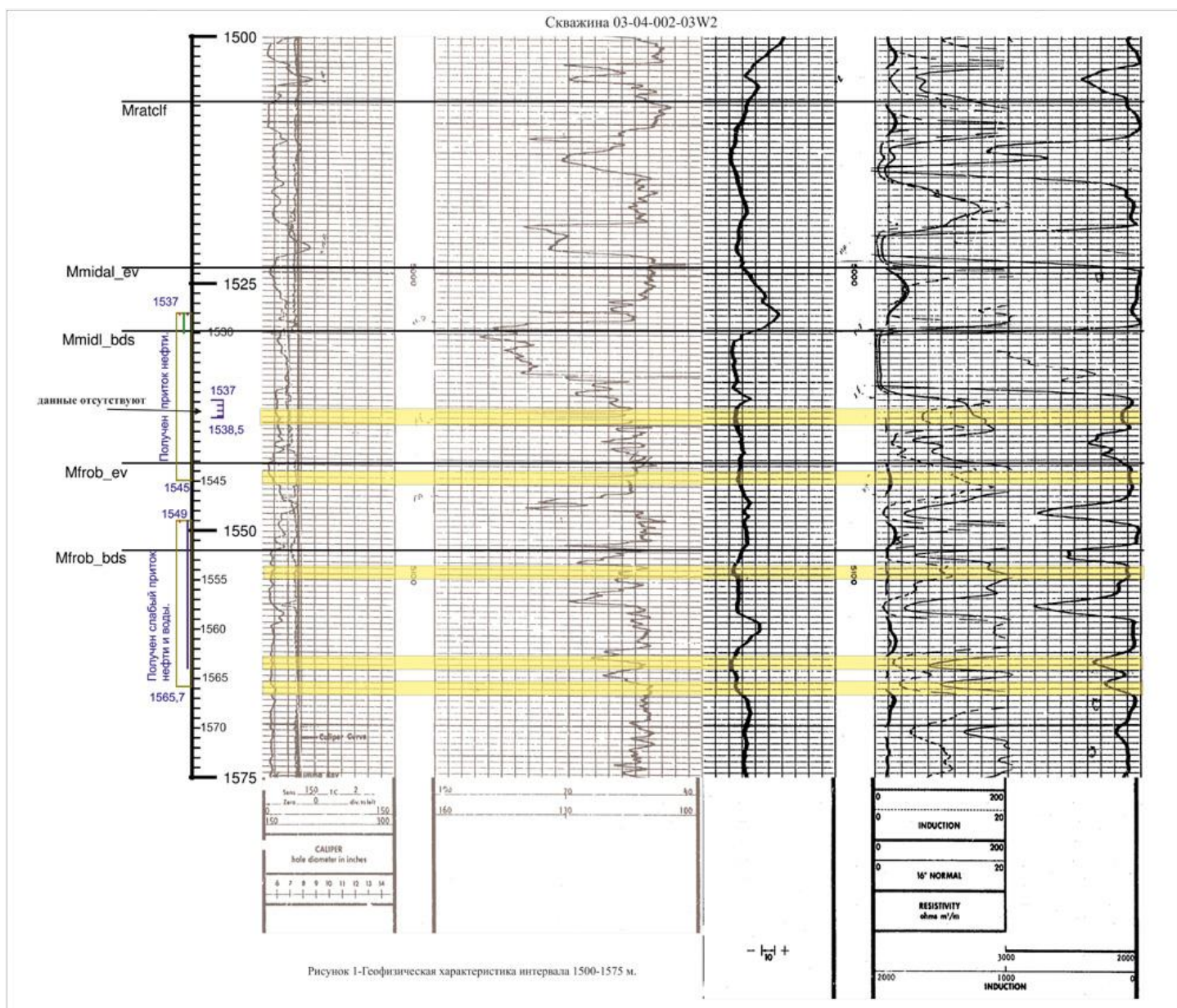
CONCLUSION

- ▶ On the basis of the submissions and comparison with the well **07-27-004-07W2** (Midale formation) for slotted perforations in the well **03-04-002-03W2** is recommended the following intervals :

Number of interval	Interval (m)	Formation	Between the cuts (m)	Between the intervals (m)
1538.0	1538.0 - 1539.5	Midale	-	-
1544.0	1544.0 - 1545.3	Frobisher	-	-
1553.5	1553.5 - 1554.8	Frobisher	-	-
1562.8	1562.8 - 1564.0	Frobisher	-	-
1565.2	1565.2 - 1566.6	Frobisher	-	-



- ▶ The casing **OD** diameter is **114.3 mm**. Internal diameter (**ID**) **103.89 mm**.
- ▶ The temperature in the production interval **57°C**. The working fluid must be heated to a temperature **30°C**.
- ▶ Specified intervals in the Frobisher formation cutting can be changed along the way. In the interval **1538.0 - 1539.5 m** not excluded the presence of water.
- ▶ Figure 1 shows the geophysical characteristics of the well **03-04-002-03W2** in the range of **1500.0 - 1575.0 m**:



- ▶ Figure 2 correlation diagram of wells **03-04-002-03W2**, **07-27-004-07W2**:



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101/03-04-002-03W2/00

121/07 27 004 07 2/00 4330

16035.9 m3
Presburg
1903/07/23
9893.5 m3
Examples C

граница фации Maastricht

В скважине 07-27-004-7W2 при опробовании интервала 1476,5 - 1480,1 м (абсолютные отметки минус 881,2-884,8м) получен приток нефти, газа и воды ($Q_{\text{г}}=0,62 \text{ м}^3/\text{сут}$; $Q_{\text{в}}=0,77 \text{ м}^3/\text{сут}$; $Q_{\text{н}}=0,05 \text{ тыс. м}^3/\text{сут}$). Дата опробования 26.05.2009 г.

рекомендуемые интервалы для перфорации

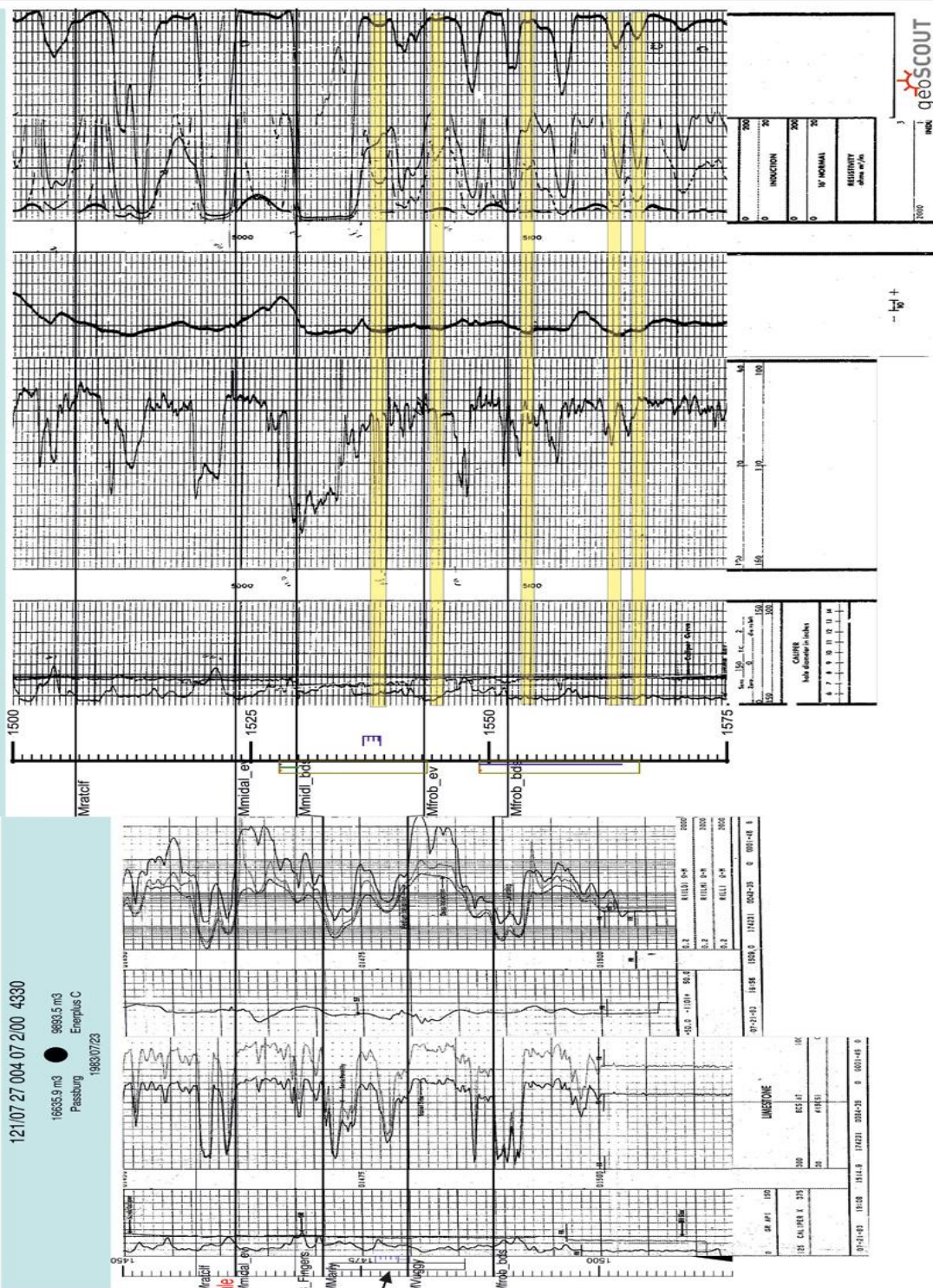


Рис. 2- Схема корреляции по скважинам 07-27-004-07 W2, 03-04-002-03W2.



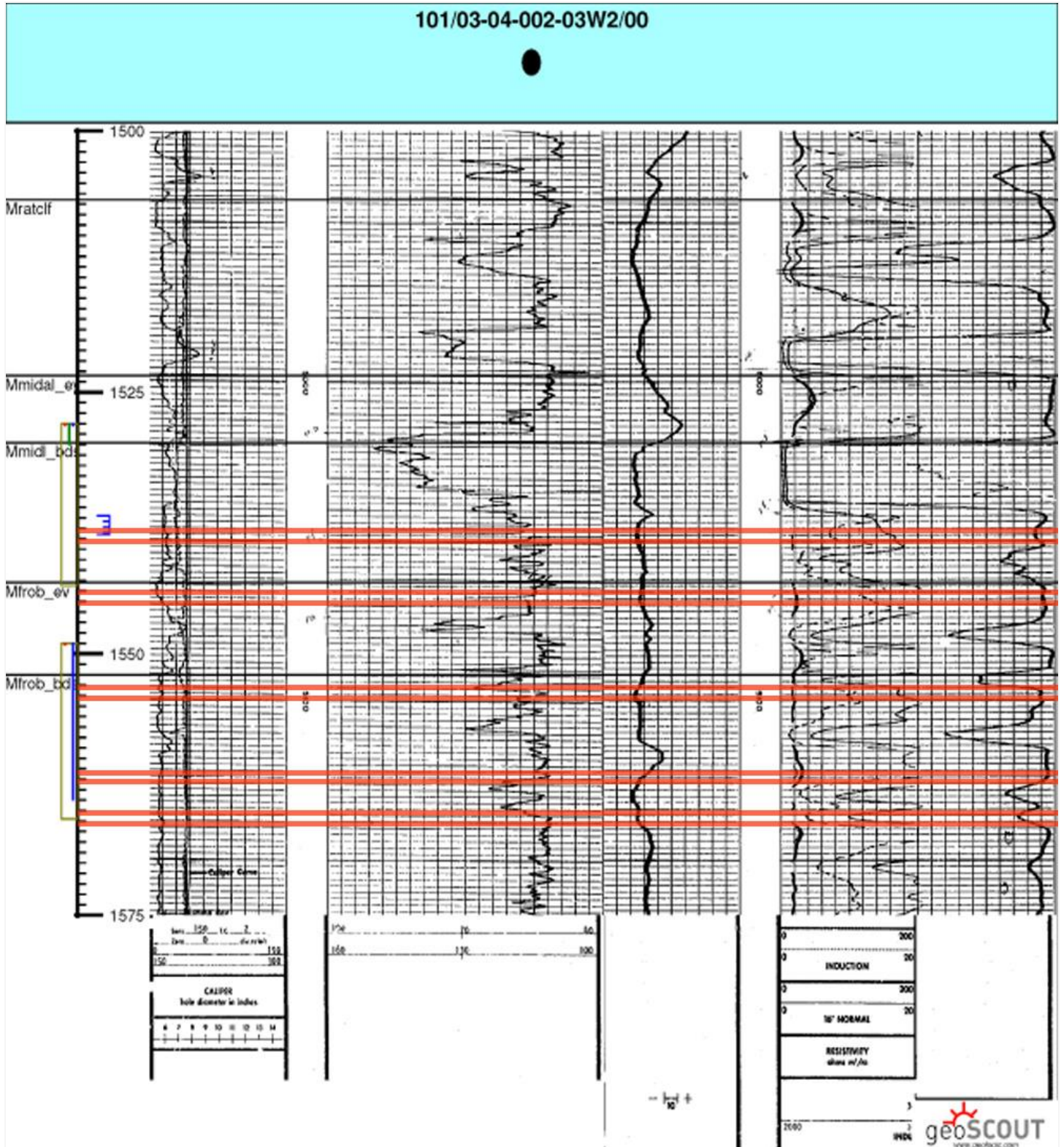
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► According to the results highlighted the following intervals for cutting. Taken into account that the upper interval is the best, it is taken fully. In the lower ranges, given that the breed is more saturated with water, taken for cutting the upper parts of these intervals:





Addition conclusion for the well Barracuda Pinto 03-04-002-03W2 (74G003).

- ▶ In the well **03-04-002-3W2 (74G003)** interval **1536.8 - 1538.6 m** according to hydrodynamic studies (**HDS**) is not characterized as a collector. The top of the collector curve by receptivity bounces at a depth **1538.0 m**. According to hydrodynamic studies (**HDS**) for slotting perforation is possible to recommend interval **1538,0-1539,5 m**.
- ▶ The collector in the interval **1530,0-1535,5 m** has low resistance by induction logging (**IL**). This interval was not tested after drilling. It was perforated interval **1537,0-1538,5 m**, but the data of the results of the influx is absent.
- ▶ If in the neighbouring well is opened similar to the interval **1530,0-1535,5 m** in the well **03-04-002-3W2 (74G003)**, and it is working without water, so in this interval is possible to conduct the slotting perforation.
- ▶ The intervals **1544,0-1545,3 m**, **1553,5-1554,8 m**, **1562,8-1564,0 m** and **1565,2-1566,6 m** in the Frobisher formation are possible additional sources of the production growth.
- ▶ It is possible to recommend intervals for the slotting perforation are shown in Figure 2. and Table 2.

Table 2.

top (m)	bottom (m)	distance (m)	saturation
1530.0	1530.4	0.4	oil
1530.6	1531.0	0.4	
1531.2	1531.6	0.4	
1531.8	1532.2	0.4	
1532.4	1532.8	0.4	
1533.0	1533.4	0.4	
1533.6	1534.0	0.4	
1534.2	1534.6	0.4	
1534.8	1535.1	0.3	
1535.2	1535.5	0.3	
total:		3.8 m	
1538.0	1538.4	0.4	oil
1538.6	1539.0	0.4	
1539.2	1539.5	0.3	
total:		1.1 m	
top (m)	bottom (m)	distance (m)	saturation
1544.0	1544.3	0.3	oil + water
1544.5	1544.8	0.3	
1545.0	1545.3	0.3	
total:		0.9 m	



CUTTING INTERVALS

Primary (upper) zone (1536.8 - 1538.7 m)

(practically continuous slot, with a distances between cuts 0.2 m)

Interval 1	1538.2 - 1538.7 m
Interval 2	1537.5 - 1538.0 m
Interval 3	1536.8 - 1537.3 m
Total intervals :	1.5 m (1.8 m)
Total time :	~ 6.0 hours

Secondary (down) zone (1529.5 - 1535.0 m)

(with a distance between cuts 0.5 m)

Interval 1	1534.5 - 1535.0 m
Interval 2	1533.5 - 1534.0 m
Interval 3	1532.5 - 1533.0 m
Interval 4	1531.5 - 1532.0 m
Interval 5	1530.5 - 1531.0 m
Interval 6	1529.5 - 1530.0 m
Total intervals :	3.0 m (5.50 m)
Total time :	~ 12.0 hours

STATISTICS

initial level	1538.2 m
final level	1529.5 m
total distance	8.7 m
total slot	1.5 m (1.8 m) + 3.0 m (5.5 m) = 4.5 m (7.3 m)
continuous slot	4.5 m (7.3 m)
gross surface drainage	~ 40.0 m ² (2 nozzles) ~ 70.0 m ² (4 nozzles)
continuous surface drainage	~ 25.0 m ² (2 nozzles) ~ 50.0 m ² (4 nozzles)

APPROXIMATE SETTINGS

Abrasive quartz sand (~ 10/35) optional (~20/40)	
2 nozzles	~ 22.0 tons
4 nozzles	~ 44.0 tons
Net cutting time	
cutting speed 3.0 mm/min	~ 25 hours 00 min
cutting speed 3.5 mm/min	~ 21 hours 30 min
cutting speed 4.0 mm/min	~ 18 hours 45 min
cutting speed 4.5 mm/min	~ 16 hours 40 min
cutting speed 5.0 mm/min	~ 15 hours 00 min
cutting speed 5.5 mm/min	~ 13 hours 40 min
Pumping settings (with 2 nozzles)	
start slurry rate	~ 0.3 m ³ /min
final slurry rate	~ 0.7 m ³ /min
working pressure	~ 27.5 - 41.5 MPa
Pumping settings (with 4 nozzles)	
start slurry rate	~ 0.6 m ³ /min
final slurry rate	~ 1.94 m ³ /min
working pressure	~ 27.5 - 41.5 MPa



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HYDRO-SLOTTING PERFORATION PROGRAM

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Program Name
Well Name
Area Name
Date
Prepared by
Prepared for

MAXXWELL PRODUCTION INC. HYDRO-SLOTTING PERFORATION TECHNOLOGY PROGRAM
BARRACUDA PINTO 03-04-002-03W2 (74G003 LEASE)
MIDALE FORMATION - BARRACUDA PINTO 03-04-002-03W2 (74G003 LEASE) - SASKATCHEWAN PROVINCE - CANADA
DECEMBER 10, 2013 (CORRECTED VERSION)
ANATOLI NIKOULINE - HYDRO-SLOTTING PERFORATION TECHNOLOGY ENGINEER (MAXXWELL PRODUCTION INC.)
FOR ADMIRALTY OILS LTD. (SASKATCHEWAN) (BARRACUDA PINTO 03-04-002-03W2 (74G003 LEASE))

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WELL & COMPLETION DATA

Casing				(Fill in one dimension cell only, inches or millimeters)				(Links for casings)			
OD Casing		inches	114.3	mm	Drift	100					
ID Casing		inches	103.89	mm							
Other parameters		pound/ft		kg/m	Steel Grade	J-55 ST&C					
Tubing				(Fill in one dimension cell only, inches or millimeters)				(Links for casings)			
OD Tubing		inches	60.33	mm	Drift	48.28					
ID Tubing		inches	50.67	mm	Internal Yield						
Other parameters		pound/ft		kg/m	Steel Grade	J-55 EUE T&C					
Cement area				(Fill in one dimension cell only, inches or millimeters)				(Links for casings)			
OD Drill Bit		inches	200.00	mm	Mud Weight Used						
Cement weight		lb/gal		kg/m ³							
Cement used					Cement Grade						
Treatment				(Fill in one dimension cell only, feet or meters)							
Total Well Depth		ft	1584.00	m	TREATMENT INTERVALS: 1529.5-1538.7 m						
Formation	MIDALE FORMATION										
Treatment interval		ft to		ft	or (l)	1529.50	m to	1538.70	m		
Gross interval		ft	9.20	m	Intervals between the cutting 20-50 cm						
Additional				(Fill in one dimension cell only)							
Open hole used		inches	200.00	mm							
Mud weight used			lb/gal		kg/m ³						
Bottom hole temp		F°	57.00	C°		C°		F°			
Casing Collars		ft		ft		m		m			
Fluid salinity	46000.00	ppm									

					(ft ³)	(m ³)	(liters)	gal (us)	bbl (us)	bbl (oil)	bbl (uk)	gal (uk)
Casing area volume (inside)	ft / or	1584.00	m		1183.39	33.51	33510	8859.1	281.0	210.8	204.7	7371.1
Tubing area volume on	ft / or	1529.50	m		154.41	4.37	4372	1155.9	36.7	27.5	26.7	961.8
Tubing area volume on	ft / or	1538.70	m		155.34	4.40	4399	1162.9	36.9	27.7	26.9	967.5
Tubing area volume (inside)	ft / or	1529.50	m		108.92	3.08	3084	815.4	25.9	19.4	18.8	678.4
Tubing area volume (inside)	ft / or	1538.70	m		109.57	3.10	3103	820.3	26.0	19.5	19.0	682.5
Fluid Volume (Well) for	ft / or	1529.50	m		528.49	14.97	14965	3956.4	125.5	94.1	91.4	3291.9
Fluid Volume (Well) for	ft / or	1538.70	m		528.22	14.96	14957	3954.3	125.4	94.1	91.4	3290.2
Average Volume (Well)	ft / or	1534.10	m		528.35	14.96	14961	3955.3	125.5	94.1	91.4	3291.0
Volume of working fluid in the surface system and tank volume (min)					1057	30	29922	7911	251	188	183	6582

SURFACE EQUIPMENT

The following surface equipment is required on site for hydro-slotting perforation technology completion:

RIG									
		(ft³)	(m³)	(liters)	gal (us)	bbl (us)	bbl (oil)	bbl (uk)	gal (uk)
1	Rig, plus tubing trailer with tubing, to be spotted opposite rig.								
2	Water storage tank volume approximately (minimum)	1057	30	29922	7911	251	188	183	6582
3	Cuttings tank volume approximately (minimum)	640	18	18127	4792	152	114	111	3987
4	Shale shaker (on top of cuttings tank); 1 pump (500-600 L/min) or 17.5-21.2 ft³/min or 132-159 gal (us) or 4.2-5.0 bbl (us).								
5	If not shaker; then 1 filter (0.125 particle size), 2 pumps (combined 1 m³/min) or 35.3 ft³/min or 264 (gal (US)/min or 8.4 bbl (US)/min								
6	30 m (98.4 ft) of low pressure tubing plus connections								
7	Standard BOP equipment – frac valve + 8 stripper rubbers for frac valve								

FRAC SERVICE

1	1 high-pressure (triplex) pump (minimum 35-41 MPa (5000-6000 psi) & 1 m ³ /min) or 35.3 ft ³ /min or 264 (gal (US)/min or 8.4 bbl (US)/min
2	1 blender (25-50 kg/m ³)
3	30 m (98.4 ft) of high pressure iron tubing, including connections & reverse flow manifold
4	Frac van

CONSUMABLES

1	20 m ³ /min (20 liters/min) or 706.3 ft ³ or 5283 gal (us) or 168 bbl (us) lease water – non-soapy brine from nearby well (look in the "Rig" table)
2	Friction reducer additive for 20 m (65.6 ft) or 3 water
3	22 tons abrasive grit* (2 nozzles) or 44 tons (4 nozzles) Program must be finalized to determine quantity. (See the next Table)
4	1 sand hopper or dump truck compatible with blender

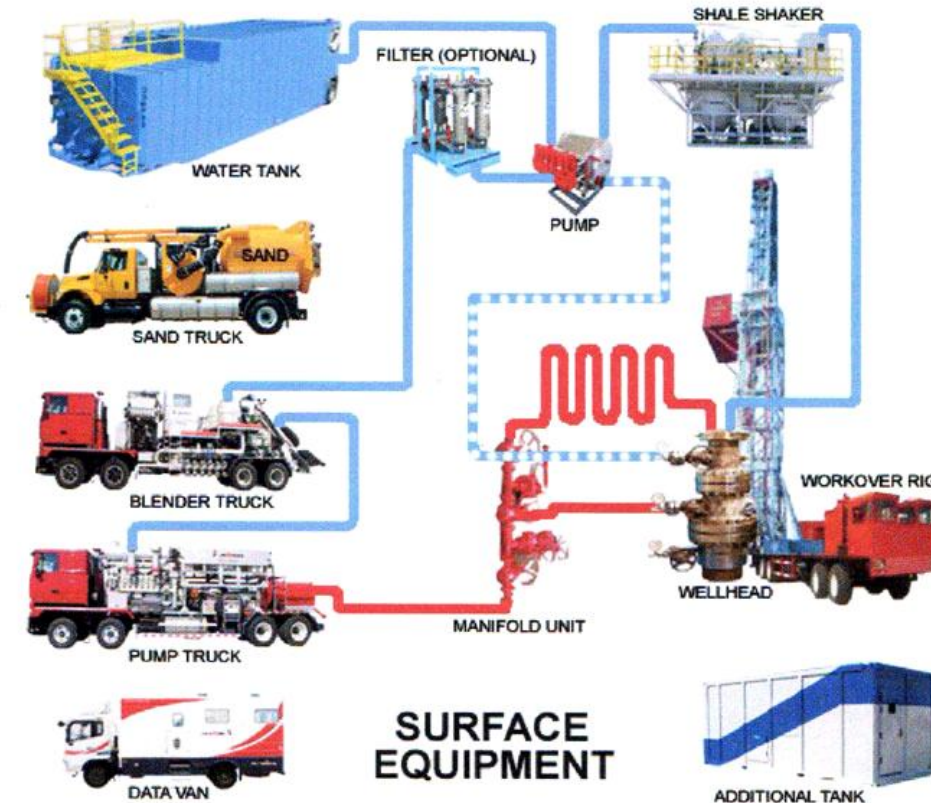


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CALCULATION THE ELONGATION OF THE TUBING UNDER THE WORKING FLUID PRESSURE, FIRST INTERVAL AND SECTIONS

A - engine length before nozzles	ft	cm	365.00
B - collar with seat length	ft	cm	80.00
C - additional pop joints section leng	ft	cm	
or			
D - full length before tubing	ft	cm	1085.00
E - marker length	ft	cm	125.00
F - one section of tubing length	ft	cm	960.00

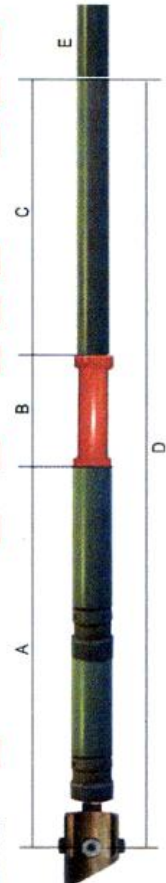
Pump performs (bbl/min)			
bbl/min	gal/min	ft ³ /min	
4.0			
Wellhead pressure (psi)			
psi	Mpa	atm	
4000.0	35.0	352.0	

Elongation of the tubing under the working fluid pressure :			
from	to		
31.3	31.1	cm	
1.03	1.02	ft	
12.31	12.26	inch	

Installation of the first interval :		
ft	m	
1538.20		

Number of sections (w/elong.):		
Start	Finish	Difference
m	m	m
1538.20	1529.50	8.70

Sections:	80	tubings
Pup joints :	0.6m (2ft)	3
Pup joints :	1.5m (5ft)	3



1	Cut 1	from		to		ft	1538.2	1538.7	m
2	Cut 2	from		to		ft	1537.5	1538.0	m
3	Cut 3	from		to		ft	1536.8	1537.3	m
4	Cut 4	from		to		ft	1534.5	1535.0	m
5	Cut 5	from		to		ft	1533.5	1534.0	m
6	Cut 6	from		to		ft	1532.5	1533.0	m
7	Cut 7	from		to		ft	1531.5	1532.0	m
8	Cut 8	from		to		ft	1530.5	1531.0	m
9	Cut 9	from		to		ft	1529.5	1530.0	m
		from	for depth	to	for depth				
Time for sand :		6.50		6.47		Average	6.49		
		min	1538.70	m	min	1529.50	m	Engine	4.0

HYDRO-SLOTTING PROCESS

SEQUENCE OF PROCEDURES

INTERVAL 1

1	Put the "tubing" ball in-to the tubing, increase the pressure	psi	6500	psi	45	Mpa	test the pressure during 1 min
2	Pressure-tested back line pressure of	psi	1500	psi	10.5	Mpa	
3	Catch the "tubing" ball with back flushing.						
4	Increase the water pressure up-to	psi	4000	psi	35	Mpa	and test tis mode during 1 min
5	Reduce the pressure and carefully start the sand	psi	3	kg/m ³	See in the Converter		and slowly increase working fluid pressure
up-to operat. mode		psi	4000	psi	35	Mpa	
Time to reach the oper. mode		min	1				
Start cutting time =		Pressure	+	Sand	=	6.3	min
Cutting time for 1 interval =						100	min
						1.8	hours
6	After	100	min	or	1.8	hours	Cut the sand (!)
7	After	6.3	min	plus	10 min	wash	Cut the pressure (!) or reduce the pressure up-to 150 psi or 1 Mpa

If possible, keep the circulation. If the transition to the next interval takes considerable time, start the back flushing.

start the reverse circulation. When the flow is restored, to continue work. If circulation flow cannot be restored switch to direct flow, lower the tool in the well and replace the slurry in the well by clean formation water. Circulate the well until no sand or debris particles are present in the flow. Catch the ball on the shaker and lift the tool from the well.

If, to maintain the pressure, it is necessary to maintain the flow, we need to stop introduction of sand and start flushing with the clean formation water, then reverse the circulation, lower the down-hole tool and replace the slurry in the well by clean formation water. Circulate the well until no sand or debris particles are present in the flow. Catch the ball on the shaker and lift the tool from the well.

The content of sand in the pulp cycle must be uniform and vary between of volume.



PREPARATION FOR HYDRO-SLOTTING PERFORATION

DAY 1

a. Prepare site. MIRU. Spot rig, spot tubing trailer opposite rig if derrick cannot stand up tubing, cuttings tank, water storage tank, other equipment according to Surface Equipment schematic. Set up surface flow lines to be able to reverse flow csg/tbg (Manifold for csg/tbg reverse flow). Vacuum trucks bring $\sim$ m³ (total: in the tank and in the well) brine and fill water storage tank. Record well pressure before taking off gauge. Bleed off any pressure (< 0.7 MPa). ND tree & NU Frac Valve. Pressure test Frac Valve. Kill well. (2 hours).

► During the process of hydro-slotting perforation it is recommended heating the water in the system (possibly steam generator). The temperature inside the well is **57°C**. For prevent the cooling of the temperature during the process the water gets into the well from the surface, recommended heating water on the surface before the temperature of the working fluid in the pipes **30°C**.

► Measure the length of the equipment assembly. During the lowering of equipment into the well to measure sections of tubing. Lower equipment to the well so that the perforator's nozzle at a depth $\sim$ **1466.0 m**.

b. (Stage 1 in the Tab) Make up and tally hydro-slotting equipment downhole assembly (equipment + transition adapter + test sub (marker) ($\sim 1 \times 1.25$ m) + pup joints ($\sim 2 \times 2$ m, 1×1 m). RIH with equipment and PU (~ 1 m) tubing from trailer & drift & tally in hole. RIH with pup joints ($\sim 2 \times 2$ m, 1×1 m).

c. (Stage 2 in the Tab) Optional RU wireline loggers. RIH with **125.0** mm gauge ring. If casing is new and no deformation is expected, and no fill is expected, well ID does not need to be.

d. Secure well & SDFN.

stage	operation (procedure)	the depth (level)	distance	approximate time
1	Lowering the tool for 1 interval	$\sim$ 1538.0 m	Elongation 30 cm	$\sim$ 2 hour 30 min
2	Correlation log			$\sim$ 3 hour 00 min
3	Correction the installation	1538.2 m		$\sim$ 0 hour 10 min
				$\sim$ 0 hour 20 min
TOTAL TIME :				$\sim$ 6 hour 00 min

HYDRO-SLOTTING PERFORATION STAGES

DAY 2

a. Warm up. RU frac equipment (pumps, blender, hopper, high-pressure lines, manifold, frac van). RU sand truck and conveyor belt to blender. Record water temperature and viscosity. Record SI pressure of tbg/csg. Test stripper rubber. Test surface lines.

► During the process of hydro-slotting perforation it is recommended heating the water in the system (possibly steam generator). The temperature inside the well is **57°C**. For prevent the cooling of the temperature during the process the water gets into the well from the surface, recommended heating water on the surface before the temperature of the working fluid in the pipes **30°C**.

b. Conduct operational and safety meeting with all personnel on location, detailing all procedures and pressure limitations.

► Hydro-slotting operations. Refer to hydro-slotting procedures (next section). Program will be comprised of 41 stages and include the following vertical intervals, in order of excavation (from deepest to shallowest) (to be varied to avoid casing collars).



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OIL WELL SLOTTING PROFESSIONAL SERVICE

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- ▶ Abrasive quartz sand (abrasive grit*) **22 tons -2 nozzles, 44 tons-4 nozzles** (abrasive quartz sand **1.5 - 3.0mm** size of grains) ~ **10/35**, optional ~**20/40**.
- ▶ During the process can be adjusted the final intervals (depth).
- ▶ Take into consideration the elongation of tubing ~ **300 mm**.
- ▶ The content of sand in the pulp cycle must be uniform and vary between 5% - 7% of volume.
- ▶ If the pressure increase over 41.5 MPa , accompanied by drop the tempo of pumping down, it is necessary to stop supply the sand, and turn reverse flushing.
- ▶ If the rate resumption - continue the process.
- ▶ If the circulation rate fails, turn reverse flushing.
- ▶ If the operating pressure drop below 10 Mpa the equipment automatic charging to its original start position (interval).

▶ Lowering up the tool

▶ Cutting process

stage	operation (procedure)	the depth (level)	distance	approximate time
4	Drop the testing ball			~ 0 hour 20 min
5	Tubing pressure testing	(within 1 min)		~ 0 hour 05 min
6	Reverse flushing			~ 0 hour 30 min
7	Catching the testing ball	(within 10 min)		~ 0 hour 10 min
8	Drop the valve (cutting) ball			~ 0 hour 20 min
9	Cutting 1 interval (cutting down)	1538.2 ▶ 1538.7 m	0.5 m	~ 1 hour 50 min
10	Direct flushing			~ 0 hour 20 min
11	Lifting UP the tool for the 2 interval	1538.2 ▶ 1537.5 m	0.7 m	~ 0 hour 05 min
12	Cutting 2 interval (cutting down)	1537.5 ▶ 1538.0 m	0.5 m	~ 1 hour 50 min
13	Direct flushing			~ 0 hour 20 min
14	Lifting UP the tool for the 3 interval	1537.5 ▶ 1536.8 m	0.7 m	~ 0 hour 05 min
15	Cutting 3 interval (cutting down)	1536.8 ▶ 1537.3 m	0.5 m	~ 1 hour 50 min
16	Direct flushing			~ 0 hour 20 min
17	Lifting UP the tool for the 4 interval	1536.8 ▶ 1534.5 m	2.3 m	~ 0 hour 10 min
18	Cutting 4 interval (cutting down)	1534.5 ▶ 1535.0 m	0.5 m	~ 1 hour 50 min
22	Direct flushing with lowering the tool from 1534.5 ▶ 1538.7 m		4.2 m	~ 0 hour 50 min
23	Lifting UP the tool on the surface with marks on the tubing.			~ 1 hour 30 min
24	Inspection the tool. Replacement nozzles, perforator or tool.			~ 1 hour 00 min
PUMPING SERVICE WORKING TIME :				~ 10 hour 45 min
RIG SERVICE WORKING TIME :				~ 12 hour 15 min
TOTAL TIME :				~ 13 hour 15 min

HYDRO-SLOTTING PERFORATION STAGES

DAY 3

stage	operation (procedure)	the depth (level)	distance	approximate time
25	Lowering the tool for 6 interval	~ 1533.5 m	Elongation 30 cm	~ 2 hour 30 min
26	Cutting 6 interval (cutting down)	1533.5 ▶ 1534.0 m	0.5 m	~ 1 hour 50 min
27	Direct flushing			~ 0 hour 20 min
28	Lifting UP the tool for the 7 interval	1533.5 ▶ 1532.5 m	1.0 m	~ 0 hour 15 min
29	Cutting 7 interval (cutting down)	1532.5 ▶ 1533.0 m	0.3 m	~ 1 hour 10 min
30	Direct flushing			~ 0 hour 20 min
31	Lifting UP the tool for the 8 interval	1532.5 ▶ 1531.5 m	1.0 m	~ 0 hour 05 min
32	Cutting 8 interval (cutting down)	1531.5 ▶ 1532.0 m	0.5 m	~ 1 hour 50 min



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33	Direct flushing			~ 0 hour 20 min
34	Lifting UP the tool for the 9 interval	1531.5 ▶ 1530.5 m	1.0 m	~ 0 hour 10 min
35	Cutting 9 interval (cutting down)	1530.5 ▶ 1531.0 m	0.5 m	~ 1 hour 50 min
36	Direct flushing			~ 0 hour 20 min
37	Lifting UP the tool for the 10 interval	1530.5 ▶ 1529.5 m	1.0 m	~ 0 hour 15 min
38	Cutting 10 interval (cutting down)	1529.5 ▶ 1534.0 m	0.5 m	~ 1 hour 50 min
39	Direct flushing			~ 0 hour 20 min
40	Direct flushing with lowering the tool from 1529.5 ▶ 1566.2 m		4.5 m	~ 1 hour 30 min
41	Lifting UP the tool on the surface with marks on the tubing.			~ 1 hour 30 min
PUMPING SERVICE WORKING TIME :				~ 12 hour 00 min
RIG SERVICE WORKING TIME :				~ 16 hour 00 min
TOTAL TIME :				~ 16 hour 00 min

ENTERING VALUES FOR THE CALCULATION OF HYDRO-SLOTTING PERFORATION

OD Casing	114.3 mm (4.5")
ID Casing	103.89 mm (4.09")
Drift Casing	~ 100.0 mm
OD Tubing	60.33 mm
ID Tubing	50.67 mm
Drift Tubing	48.28 mm
OD Drill Bit	200.0 mm
Cement weight	1400 kg/m
Cement used	SpectraCem NP + 0.9% FL-5 + 0.8% LCB-1000
Total Well Depth	1584.0 m
Formation	Midale / Frobisher formations
Treatment interval	1529.5-1535.0 m and 1536.8-1538.7 m
Continuous interval	4.5 m
Open hole used	200.0 mm
Mud weight used	1100 to 1300 kg/m ³
Bottom hole temp	57 °C
Casing Collars	-
Fluid salinity	46000 ppm

RESULTS

Casing area volume (inside) 1584.0 m	33.51 m ³
Tubing area volume on 1529.5 m	4.37 m ³
Tubing area volume on 1538.7 m	4.40 m ³
Tubing area volume (inside) 1529.5 m	3.08 m ³
Tubing area volume (inside) 1538.7 m	3.10 m ³
Fluid Volume (Well) for 1529.5 m	14.97 m ³
Fluid Volume (Well) for 1538.7 m	14.96 m ³
Average Volume (Well) 1534.1 m	14.96 m ³
Pump performers	0.3-0.7 m ³ /min (2 nozzles) 0.6 - 1.94 m ³ /min (4 nozzles)
Working pressure	27.5 - 41.5 MPa
Equipment speed	4.0 - 4.5 - 5.0 (corrected during the operation)
Full length of the equipment (before tubing)	3.4 m
Marker length	~ 1.25 m
One section of tubing length	~ 9.6 m
Installation of the first interval	1538.2 m (With elongation= 30 cm= 1508.2 m)
Elongation of tubing (~ 30 MPa)	300 mm



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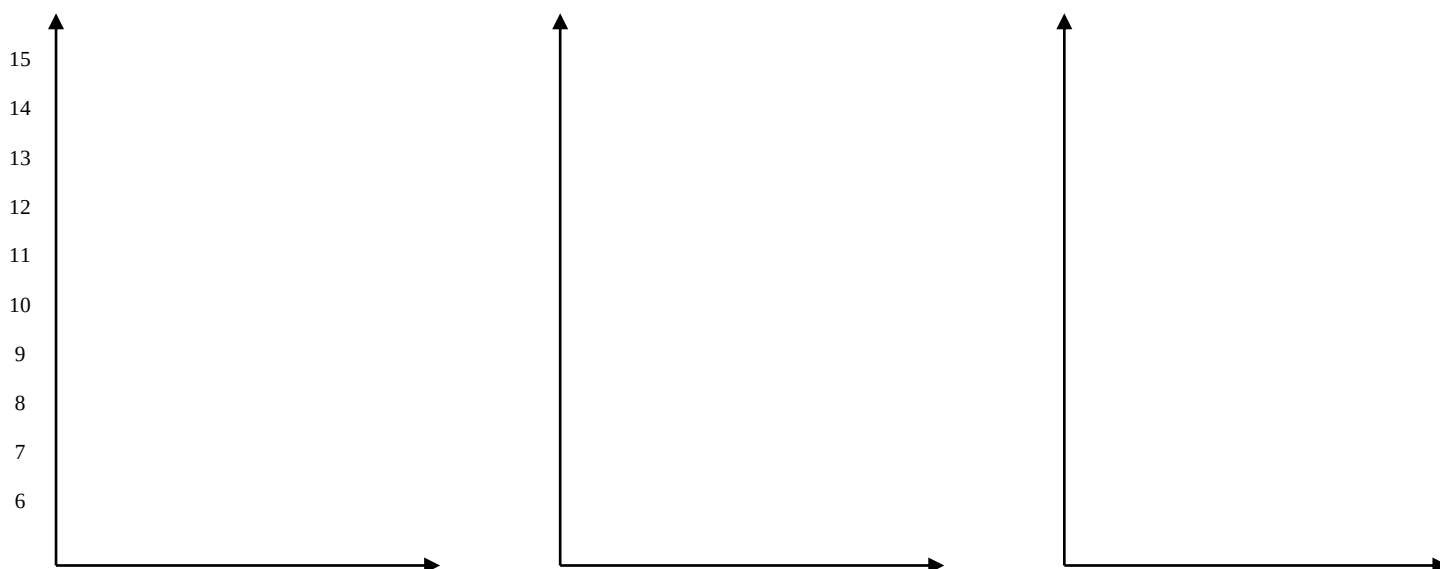
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Time to reach the sand to nozzles	~ 6 min 50 sec
Time to reach the sand to surface	~ 18 min 30 sec
Pressure loss in tubing	~ 2.0 MPa
Pressure drop across the nozzles	~ 35.3 MPa
Friction coefficient	~ 1.5
Pup joints	0.6 m + 1.5 m
Volume of working fluid in the surface system	30 m ³
Water storage tank volume approximately (minimum)	20 m ³
Cutting tank volume approximately (minimum)	20 m ³
Shale shaker (on top of cuttings tank)	1 pump (500-600 l/min)
If not shaker; then 1 filter	(0.125 particle size) 2 pumps (combined 1 m ³ /min)
Low pressure tubing plus connections	30 m (98.4 ft)
Standard BOP equipment	Frac valve + 8 stripper rubbers for frac valve
1 high-pressure (triplex) pump	minimum 35 MPa & 1 m ³ /min
1 blender	25-50 kg/ m ³
High pressure iron tubing & reverse flow manifold	30 m (98.4 ft) including connections
Frac van	1
Non-soapy brine from nearby well	50 m ³ (46 m ³)
Friction reducer additive for	20 m (65.6 ft) or 3 water
Abrasive quartz sand (abrasive grit*)	40 tonnes (abrasive quartz sand 1.5-3.0 mm size of grains)

LOSS PRESSURE

WORKING PRESSURE

WELLHEAD PRESSURE



POSSIBLE CHEMICAL TREATMENT

Continue completion, possible chemical treatment, and test well.

The approximate composition:

Acid:	15% HCl
Surface-active substances:	0.3%-0.5% (water-repellent agent for for debit wells)
Corrosion inhibitor:	0.5%
Stabilizer:	0.1%
Moderator of the reaction rate:	0.1% (in US three ready types).



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Solution volume is calculated from the drainage radius equal 5 feet from the wellbore.

